



COMSATS University Islamabad
Department of Computer Science (BSAI-V)
SPRING-2026

CSC 232 – Information Security

Due Date :22/5/2026

**CLO5 - Develop a project addressing a real-world problem in
cybersecurity in a team environment.**

Authentication Tasks

1. Username enumeration via different responses
2. Broken brute-force protection, IP block
3. 2FA bypass using a brute-force attack

Access Control Tasks

1. Insecure direct object references (IDOR) – Account page
2. Access control vulnerabilities in multi-step processes
3. Access control bypass via Referer header

Machine 1: TryHackMe – “Pickle Rick”

Tasks

1. Scan the machine using Nmap and identify open ports.
2. Find hidden files on the web server.
3. Identify the username and password hidden inside files.
4. Log in to the portal using discovered credentials.
5. Get a reverse shell on the machine.
6. Find all 3 ingredients (flags) from the machine.
7. Explain how you achieved privilege escalation (very simple, 1 command).

Hints

- Use gobuster for directory brute-forcing.
- Look for .txt and .php files.
- Use simple Python reverse shell.

Learning Outcomes

- Basic web enumeration
- Reading hidden files
- Reverse shell basics
- Simple privilege escalation

Machine 2: Hack The Box – “Blue”

Tasks

1. Run full Nmap scan and detect the target vulnerability.
2. Identify if the machine is vulnerable to MS17-010 (EternalBlue).
3. Use Metasploit to exploit the machine.
4. Get a Meterpreter shell on the target.
5. Extract system information from the shell.
6. Locate the flag on the machine.

Hints

- Search exploit in Metasploit: search ms17_010
- Set the correct RHOSTS value
- Exploit → shell comes automatically

Learning Outcomes

- Real-world vulnerability exploitation
- Using Metasploit safely
- OSCP-style structured approach

Machine 3: TryHackMe – “RootMe”

Tasks

1. Run Nmap to identify open ports.
2. Enumerate the website and find upload functionality.
3. Bypass upload filter to upload a reverse shell.
4. Get a reverse shell on the machine.
5. Enumerate SUID binaries.
6. Use one SUID binary to get root access.
7. Capture user.txt and root.txt flags.

Hints

- PHP reverse shell is easiest.
- Use `find / -perm -4000 -type f 2>/dev/null`
- `python -c` command is useful.

Learning Outcomes

- File upload vulnerability
- SUID privilege escalation
- Post-exploitation fundamentals

Machine 4: TryHackMe – “Simple CTF”

Tasks

1. Identify open ports using Nmap.
2. Try FTP login with common credentials.
3. Enumerate the website for hidden directories.
4. Enumerate Wordpress (if detected).

5. Gain shell access on the machine.
6. Perform basic privilege escalation.
7. Capture both flags.

Hints

- Anonymous FTP might be enabled.
- Use wpscan for Wordpress.
- Check sudo -l.

Learning Outcomes

- FTP misconfiguration
- CMS enumeration
- Simple privilege escalation
- Web application weakness discovery